Lukas Helmig

PhD Student in Theoretical Physics

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Education

PhD **Osnabrück University**, Physics — Osnabrück, Germany
- Investigating mixing kinetics in disordered systems. — 2025 – 2028

MS **University of Cologne**, Physics — Cologne, Germany
- Specialized in solid state theory and quantum information. — 2022 – 2025
- Thesis on "Restricted Boltzmann Machines and Hopfield Networks".
- Top of the year.

Ecole Normale Supérieure de Paris, Physics — Paris, France
- Participated in the Erasmus exchange program, focusing on advanced topics in theoretical physics. — 2023 – 2024

BS **Osnabrück University**, Physics — Osnabrück, Germany
- Thesis on "Influence of interactions on the efficiency at maximal power of a microscopic thermodynamic machine". — 2018 – 2021
- Top of the year. Finished with honors.

Experience

Faculty of Physics, Osnabrück University, Research Assistant — Osnabrück, Germany
Worked on the VRnano project, a physical development environment for better control of laboratory experiments. — 2021 – 2023 (2 years)

Faculty of Mathematics, Osnabrück University, Teaching Assistant — Osnabrück, Germany
Tutorial lectures and exam grading. — 2019 – 2021 (2 years)

Awards

Honors branch — 2022
Awarded for outstanding academic achievement
University of Cologne

Niedersachsen Scholarship — 2021
Awarded for outstanding academic achievement
Niedersachsen Ministry of Science and Culture

Deutschlandstipendium — 2019
Awarded for outstanding academic achievement from 2019 to 2021
Osnabrück University

Teaching

Theoretical Physics II, Teaching Assistant — Osnabrück, Germany
- Introduction to quantum mechanics and statistical mechanics. — 2026 – 2027
- Held weekly exercise classes and provided feedback to students.

Pre-course in Mathematics, Lecturer — Osnabrück, Germany, 2026
- Taught a 2-week pre-course in mathematics for first-year physics students.
- Covered topics such as linear algebra, calculus, and probability theory.

Theoretical Physics III, Teaching Assistant — Osnabrück, Germany, 2026
- Advanced topics in classical mechanics and electrodynamics.
- Held weekly exercise classes and provided feedback to students.

Theoretical Physics IV, Teaching Assistant — Osnabrück, Germany, 2025 – 2026
- Advanced topics in quantum mechanics and statistical mechanics.
- Held weekly exercise classes and provided feedback to students.

Outreach

DOPPLERS, Volunteer — Osnabrück, Germany, 2026
DOPPLERS is a yearly theoretical physics competition held at different german universities and supported by the German Physical Society (DPG).
- Organized the academic program including two speakers and held a workshop on decision-making myself.
- Sought and acquired sponsoring for the catering.
- Volunteered and helped with the event in general.

Deutsche Schülerakademie, Course Instructor — Jülich, Germany, 2025
The course was held in Jülich, Germany, and was part of the Deutsche Schülerakademie program.
- Developed a 2-week course (50 hours) on quantum computing for high school students.
- Conducted lectures and exercise lessons
- Guided students through a final project phase with an accompanying poster session.